

Cloud-Native Telecom Intelligence Platform

with AI Operations Agent (HCS-Ready)

End-of-Studies Project (PFE) Proposal

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Chapter 1

Executive Summary

Modern telecom networks generate massive operational data, yet the true business impact often remains hidden — this project transforms raw performance signals into actionable intelligence through cloud-native AI.

This project proposes the design and implementation of a cloud-native telecom intelligence platform integrating artificial intelligence for operational analytics and SLA risk evaluation.

The system simulates realistic OSS (Operational Support Systems) and BSS (Business Support Systems) environments using structured synthetic telecom data and demonstrates how modern cloud architectures can enhance visibility, automation, and decision-making processes.

The platform is modular, containerized, and architected for portability toward Huawei Cloud Stack (HCS). It combines:

- A three-layer data lake architecture
- AI-based anomaly detection
- SLA risk scoring
- OSS-BSS correlation analysis
- REST-based service exposure

The objective is not to replace vendor-grade telecom systems, but to deliver a deployable cloud-native reference architecture demonstrating AI-enhanced operational intelligence.

Chapter 2

Problem Context

Modern telecom networks generate massive volumes of operational and business data. However, operational analytics (OSS) and business analytics (BSS) are often processed independently.

2.1 Operational Support Systems (OSS)

Monitor network performance through KPIs such as:

- Latency
- Throughput
- Packet loss
- Availability

2.2 Business Support Systems (BSS)

Handle commercial and customer-related data:

- Usage records
- Billing data
- Revenue streams
- Customer activity indicators

Despite the strong interdependency between network performance and financial impact, integrated intelligence layers remain limited, especially in cloud-native architectures.

Chapter 3

Project Objectives

The project aims to design and implement a cloud-native platform that:

1. Ingests synthetic OSS and BSS telecom data.
2. Structures data using a three-layer data lake model.
3. Applies AI models for anomaly detection.
4. Computes SLA risk scoring.
5. Performs statistical OSS-BSS correlation.
6. Exposes insights via REST APIs.
7. Is deployable locally via Docker.
8. Is architected for Huawei Cloud Stack portability.

Chapter 4

System Architecture

4.1 Architectural Principles

- Microservices-based design
- Containerized deployment
- Modular and scalable components
- Separation of compute and storage
- Cloud portability (HCS-ready mapping)

4.2 Core Components

4.2.1 API Gateway

- REST interface
- External access layer
- Query exposure for KPIs and analytics

4.2.2 Pipeline Worker

- Synthetic data generation
- Data ingestion and transformation
- Data lake orchestration
- Correlation computation

4.2.3 AI Service

- Network anomaly detection
- Revenue anomaly detection

- SLA risk scoring
- Model lifecycle management

4.2.4 Storage Layer

- Object Storage (MinIO locally, OBS in HCS)
- PostgreSQL for metadata and serving layer

Chapter 5

Data Architecture

The system follows a structured three-layer data lake model.

5.1 Raw Layer

Immutable synthetic OSS and BSS data.

5.2 Processed Layer

Cleaned, aligned, and feature-engineered datasets.

5.3 Curated Layer

Analytics-ready datasets including:

- Joined OSS-BSS datasets
- AI inference outputs
- SLA risk results
- Correlation metrics

Chapter 6

Artificial Intelligence Modules

6.1 Network Anomaly Detection

Detect abnormal:

- Latency spikes
- Packet loss bursts
- Throughput degradation

6.2 Revenue Anomaly Detection

Identify abnormal revenue or usage patterns.

6.3 SLA Risk Scoring

Estimate the probability of SLA violations using rolling KPI windows and trend analysis.

6.4 OSS-BSS Correlation Engine

Analyze statistical relationships between:

- Network performance indicators
- Revenue fluctuations

Chapter 7

Deployment Strategy

7.1 Local Deployment

- Docker
- Docker Compose
- Isolated container networking

7.2 Huawei Cloud Stack Mapping

Local Component	HCS Equivalent
Docker Containers	ECS Instances
MinIO	OBS
PostgreSQL	RDS
Internal Network	VPC
API Gateway	ELB + ECS
AI Service	ECS / CCE
Pipeline Worker	ECS / CCE Job

Chapter 8

Innovation and Industrial Value

8.1 Academic Contribution

- Applied AI in telecom domain
- Cloud-native data architecture implementation
- Production-style microservice structure

8.2 Industrial Relevance

- Aligns with AI-driven operations trends
- Demonstrates cloud-native telecom blueprint
- Provides HCS-ready architecture mapping

Chapter 9

Project Scope Clarification

This project:

- Is a cloud-native reference architecture
- Uses synthetic telecom data
- Demonstrates AI-enhanced operational analytics

This project does not:

- Replace vendor-grade OSS/BSS systems
- Compete with full-scale telecom assurance suites
- Use real production data

Chapter 10

Conclusion

The proposed platform demonstrates how telecom operational intelligence can be structured using modern cloud-native architectures and AI-based analytics. It provides academic depth, industrial alignment, and a deployable technical implementation within a realistic five-month PFE scope.